

Lab Experiment 7

Breadth First Search (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal in a graph.

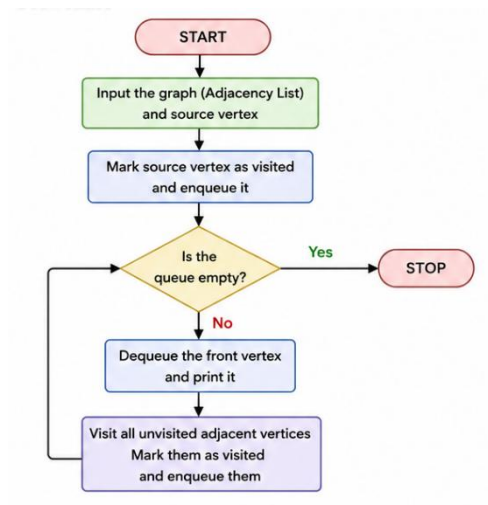
Objective

The objective of this experiment is to implement the Breadth First Search algorithm for graph traversal. It aims to visit all vertices level by level starting from a given source vertex.

Algorithm

1. Create a graph using an adjacency list.
2. Select the source vertex and mark it as visited.
3. Insert the source vertex into a queue.
4. Remove a vertex from the front of the queue and display it.
5. Visit all unvisited adjacent vertices.
6. Mark each adjacent vertex as visited and add it to the queue.
7. Repeat Steps 4–6 until the queue becomes empty.

Flowchart



Code

```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
def bfs(graph, start):  
    visited = set()  
    queue = deque([start])  
  
    visited.add(start)
```

```

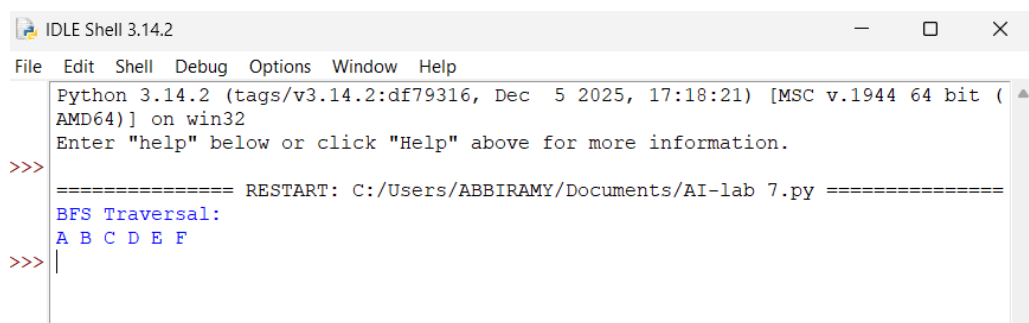
while queue:
    vertex = queue.popleft()
    print(vertex, end=" ")

    for neighbor in graph[vertex]:
        if neighbor not in visited:
            visited.add(neighbor)
            queue.append(neighbor)

print("BFS Traversal:")
bfs(graph, 'A')

```

Output



```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 7.py =====
BFS Traversal:
A B C D E F
>>> |

```

Result

The Python program successfully implemented the Breadth First Search (BFS) algorithm and traversed all the vertices of the graph level by level.

Conclusion

Breadth First Search (BFS) is an efficient graph traversal algorithm that visits vertices level by level using a queue. It is widely used for finding the shortest path in unweighted graphs.